AI Friday Season 2 – Regional Finale

AI-Powered Banking Customer Query Resolution and Fraud Alert System

PROBLEM STATEMENT

Banking customer service departments handle millions of inquiries daily, ranging from account information to transaction disputes and fraud alerts. Customers often face long wait times and inconsistent responses due to fragmented systems and manual case routing. Fraud detection teams struggle to prioritize alerts efficiently, leading to delayed responses that increase financial losses and customer dissatisfaction. Current chatbot solutions lack the ability to intelligently route complex queries or integrate real-time fraud data from multiple sources, limiting proactive risk mitigation. The scale of daily transactions and customer interactions amplifies the operational burden and compliance risks. An integrated AI solution that understands customer intent, automates fraud alert prioritization, and provides seamless multi-channel support is critical. This would reduce operational costs, enhance customer trust, and strengthen regulatory compliance by delivering timely, accurate, and personalized assistance.

Data Considerations:

Transactional data in JSON/CSV formats from internal banking systems; customer profiles from CRM APIs; real-time fraud alert feeds via REST APIs; historical customer queries and chat logs for conversation modeling; compliance guidelines datasets; data volume expected in the range of hundreds of thousands of records; preprocessing includes anonymization and normalization; privacy compliance with GDPR and PCI DSS must be ensured; synthetic data generation tools can simulate customer queries and fraud scenarios for testing.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

SOLUTION EXPECTATIONS

A web-based AI assistant featuring multi-agent conversational modules that categorize and route customer queries intelligently, integrated with real-time fraud detection APIs to prioritize alerts. Key deliverables include an advanced UI with a supervisor dashboard showing alert statuses and resolution metrics, authentication for secure access, and multi-source data aggregation. Success metrics measured by query resolution accuracy, fraud response time reduction, and system uptime above 99%. Demonstrations will showcase real-time interaction flows, alert prioritization, and multi-agent collaboration resolving complex customer issues efficiently.

Feel free to explore creative solutions that align with the problem context and objectives.

Use the following checklist as applicable while solving the selected problem statement.

USER EXPERIENCE AND INTERFACE

- Design an intuitive interface aligned to the target workflow, user journey, context, tone, and accessibility needs.
- Use conversation, visualization, or a hybrid experience where it improves clarity, personalization, and speed of action.

DATA ARCHITECTURE AND PROCESSING

- Use relevant, high-quality, anonymized or synthetic data with clear preprocessing, retrieval, aggregation, and correlation methods.
- Choose suitable storage for structured and unstructured data, ensuring consistency across sources, generated datasets, and downstream outputs.

CORE AI SOLUTION DESIGN AND AGENTIC ARCHITECTURE

- Define the agentic design pattern, agent roles, collaboration model, handoffs, tool usage, retry, reflection, and escalation logic.
- Balance autonomy with human approval checkpoints, intervention points, and clear boundaries for high-risk or uncertain decisions.
- Ground outputs in trusted sources, cite evidence where useful, separate facts from assumptions, and explain reasoning, confidence, limits, and trade-offs.
- Build responsible AI guardrails for privacy, bias, explainability, prompt injection, unsafe recommendations, unauthorized tool execution, and regulated decisions.

TECHNICAL IMPLEMENTATION

- Use a modular, scalable architecture that clearly separates AI layers, enterprise systems, knowledge stores, APIs, tools, and external feeds.
- Define context engineering across memory, caching, retrieval, prompts, context windows, token usage, latency, cost, and privacy.
- Maintain configurability across models, prompts, tools, APIs, vector stores, and deployment environments to avoid lock-in.
- Cover security, identity, permissions, secrets, data retention, audit logging, access governance, and misuse monitoring.

TESTING AND QUALITY ASSURANCE

- Evaluate beyond demo success: accuracy, groundedness, hallucination control, latency, cost, reliability, traceability, safety, and user trust.
- Use varied test data, edge cases, failure scenarios, expected outputs, scoring criteria, and automated checks where practical.
- Validate robustness, accuracy, and reliability across key use-case conditions and user journeys.

DEMO READINESS

- Prepare realistic demo flows using quality input data, edge cases, failure paths, and expected outcomes.
- Show how AI, GenAI, or Agentic AI improves the solution versus a conventional approach in speed, scale, quality, reliability, or personalization.
- Clearly explain business value, adoption path, measurable outcomes, design choices, alternatives, trade-offs, and prototype-to-enterprise readiness.
- Walk through problem breakdown, data preparation, architecture, agent roles, prompts, tools, evaluation approach, code flow, and AI-assisted code generation.